

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

MECHANICAL PROPERTIES OF FLUIDS

LECTURE-07 **PHYSICS**

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

- 1 Revision
- 2 Capillarity
- 3 .. *critical velocity*
- 4 ..
- 5 ..

RECAP FROM 1ST LECTURE

Fluid

liquid

gas

Density ρ

$$\rho = \frac{m}{V}$$

Pressure

$$P = \frac{F}{A}$$

Atmospheric pressure P_0

Variation of pressure with depth



$$P_B = P_A + \rho gh$$

Pascal Law

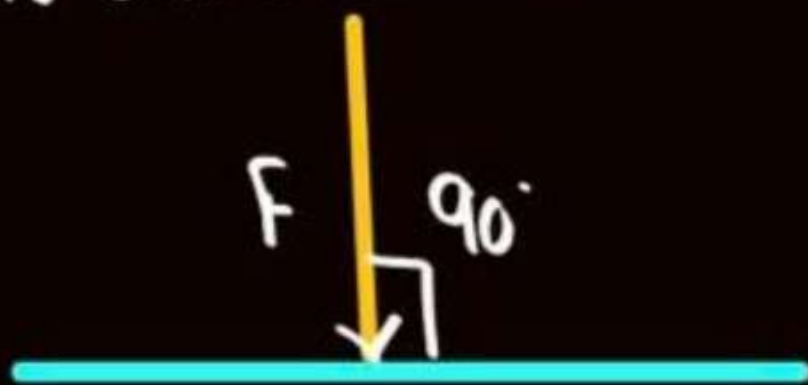
Recap-2

① Application pascal law

- Hydraulic Brake ✓
- Hydraulic lift

② Thrust

- Any perpendicular force is called thrust

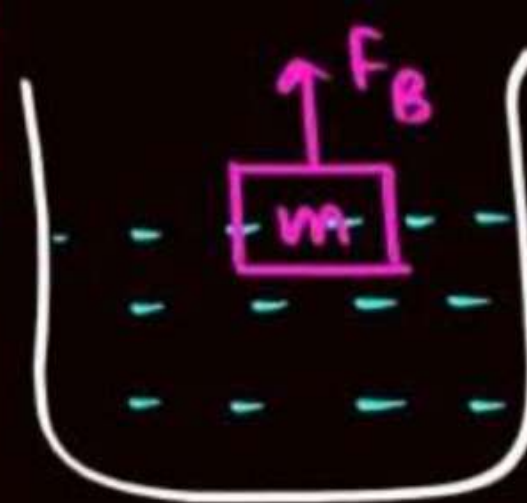


Buoyant force



Archimedes principle

$$F_B = \rho V g$$



Flow

Streamline flow

$$Re < 2000$$

Equation of Continuity

$$a_1 v_1 = a_2 v_2$$

turbulent flow

$$Re \gg \gg 2000$$

Recep - ③

Bernoullies Theorm

- law of conservation of Energy

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$$

Application of B-Theorm

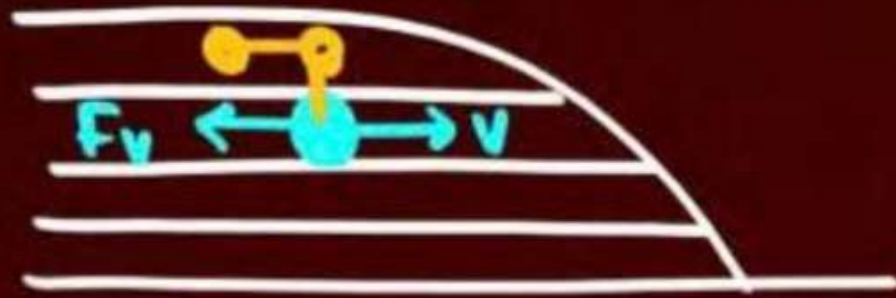
① Spray / Atomizer

② Magnus effect

③ lift of aeroplane

RECAP FROM 4TH LECTURE

• Viscosity



$$F_v = -\eta A \frac{dv}{dx}$$

$\eta \Rightarrow$ coefficient of viscosity

η = Pascal sec (SI)

η = poise (c.g.s)

Dimensional formula of η

$$\checkmark \eta = m l^{-1} t^{-1}$$

Stoke law

$$F_v = 6\pi\eta r v$$

$$a=1$$

$$b=1$$

$$c=1$$

RECAP FROM 5TH LECTURE

- Terminal velocity



$$v_T = \frac{2}{9} \frac{r^2 g (\rho - \rho_f)}{\eta}$$

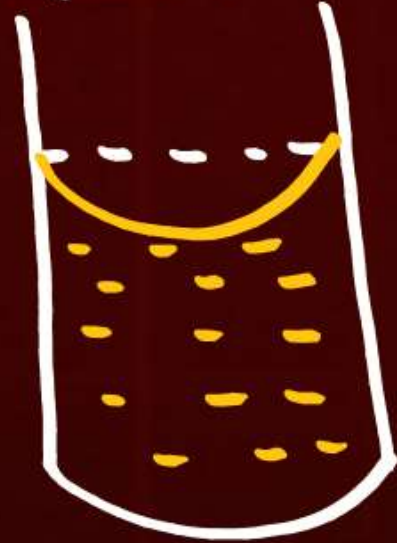
RECAP FROM 6TH LECTURE

Application of Surface tension

(a) Shape of drop



(b) Shape of water surface



Note:

Angle of contact

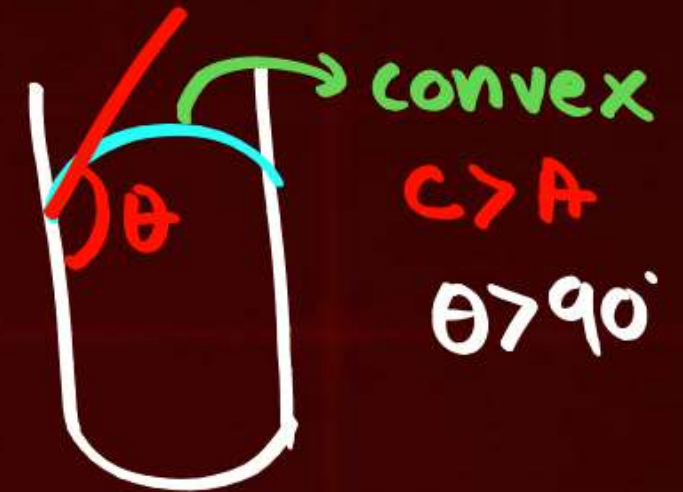
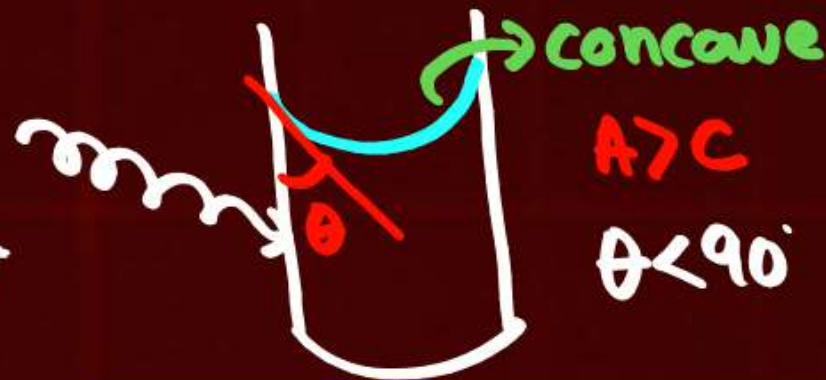
Adhesion

water + wall

Cohesion

water + water

Miniscus



Excess Pressure

$$P = P_0 + \rho g h$$

effect of $\uparrow$ g on Pressure

drop

$$\text{drop } \epsilon_p = \frac{2T}{R}$$

$$\text{bubble } \odot = \frac{4T}{R}$$

Numerical.

lecture no. 7

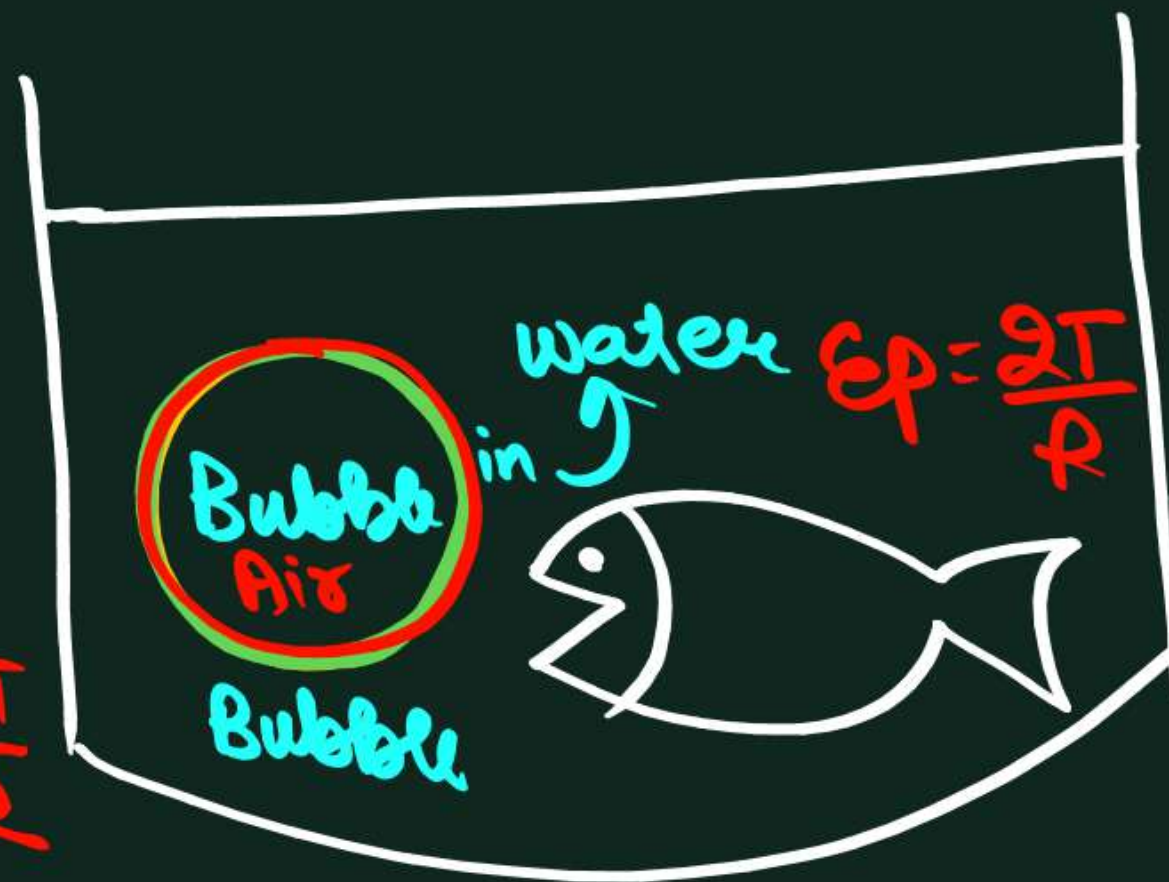
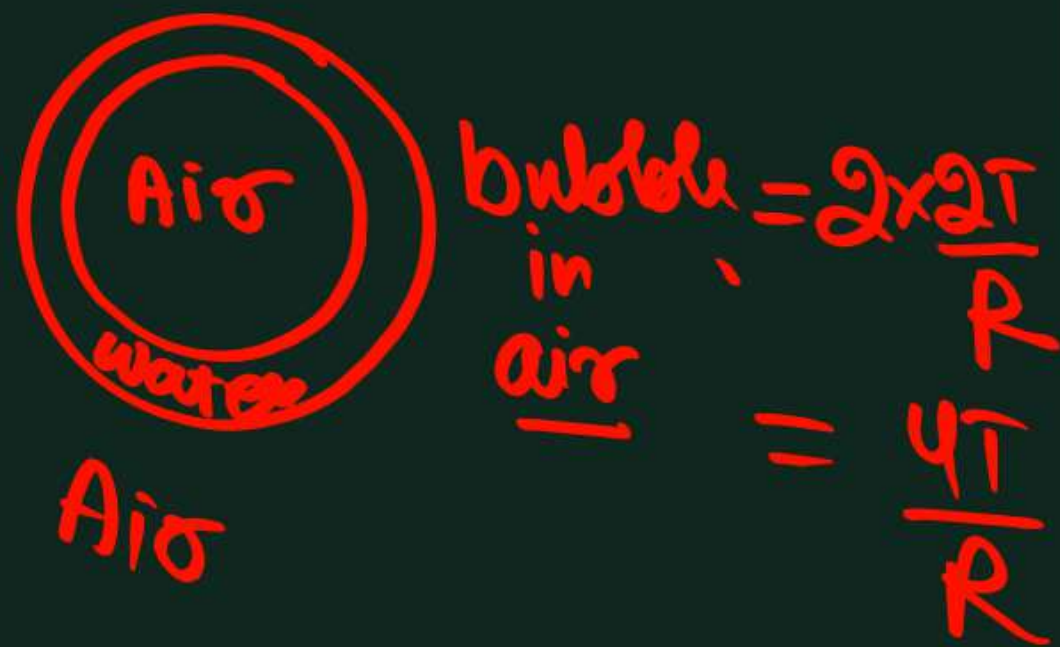
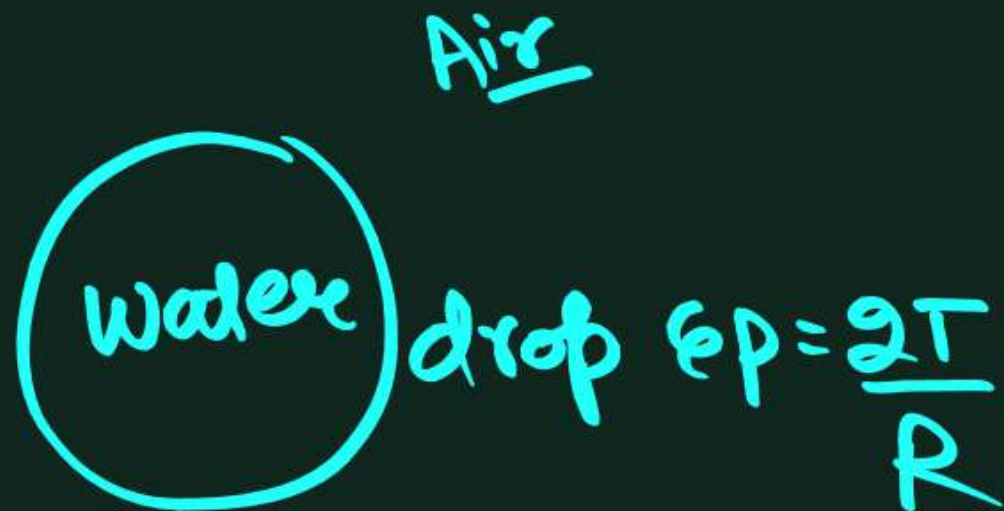
Q1 Excess pressure of bubble in water (Neet-2021 | Jee 2024) ①

(a) $\frac{2T}{R}$

(b) $\frac{3T}{R}$

(c) $\frac{4T}{R}$

(d) $\frac{5T}{R}$



Q.1) What is the excess pressure in miniscus of water

2

- (a) $\frac{2T}{R}$
(b) $\frac{3T}{R}$
(c) $\frac{4T}{R}$
(d) $\frac{5T}{R}$



Note :- ① Excess pressure depend only on number of surface

$S_1 \quad \epsilon p = \frac{2T}{R}$

$S_2 \quad \epsilon p = \frac{4T}{R}$

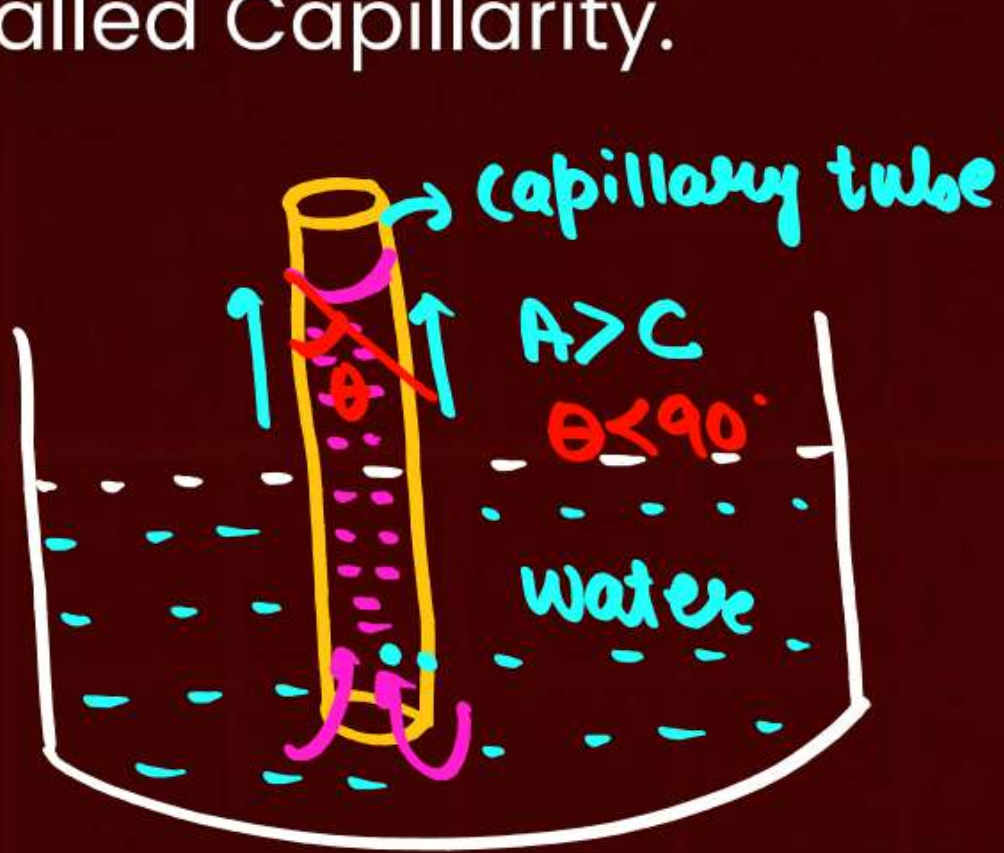
② Excess pressure = $P_0 - P_B$
 $= P_B - P_0$

Capillarity



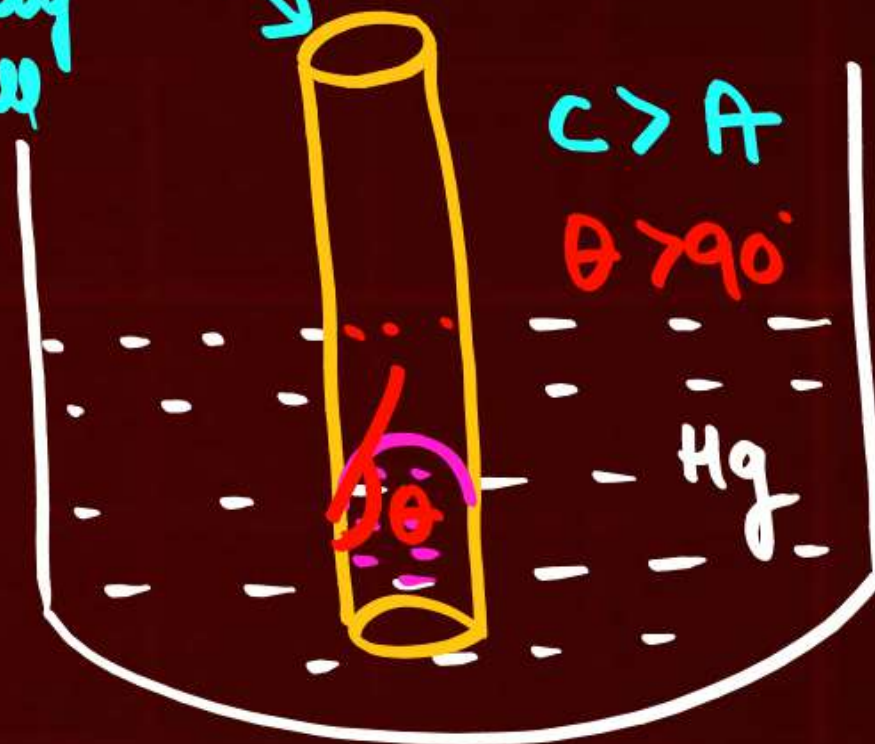
The rise or fall of a liquid in a capillary tube immersed partially in the liquid is called Capillarity.

(i) Capillary Rise
✓
5 mark



Concave Surface

(ii) Capillary Fall
✗

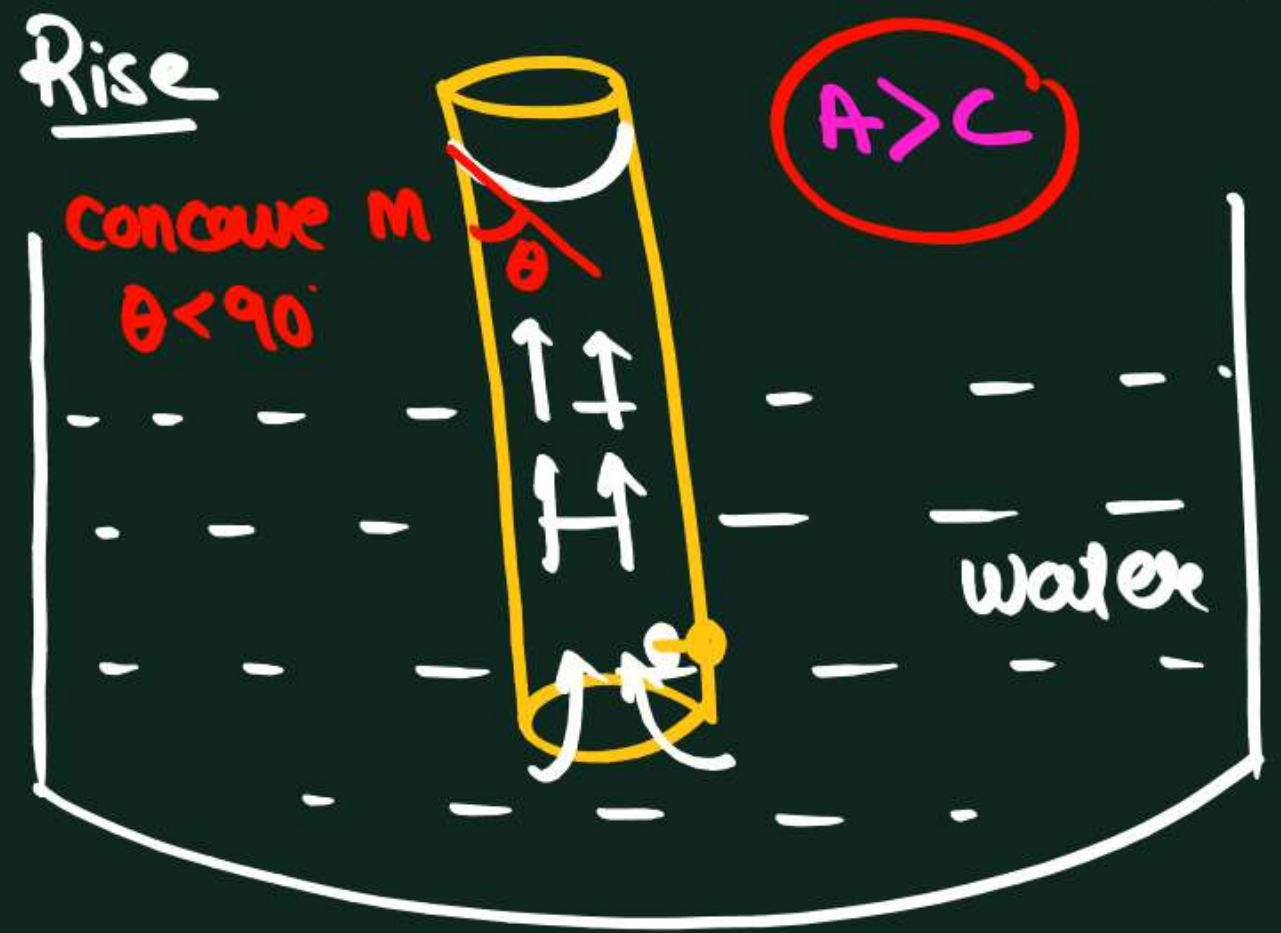


Convex meniscus.

(i) Capillary Rise

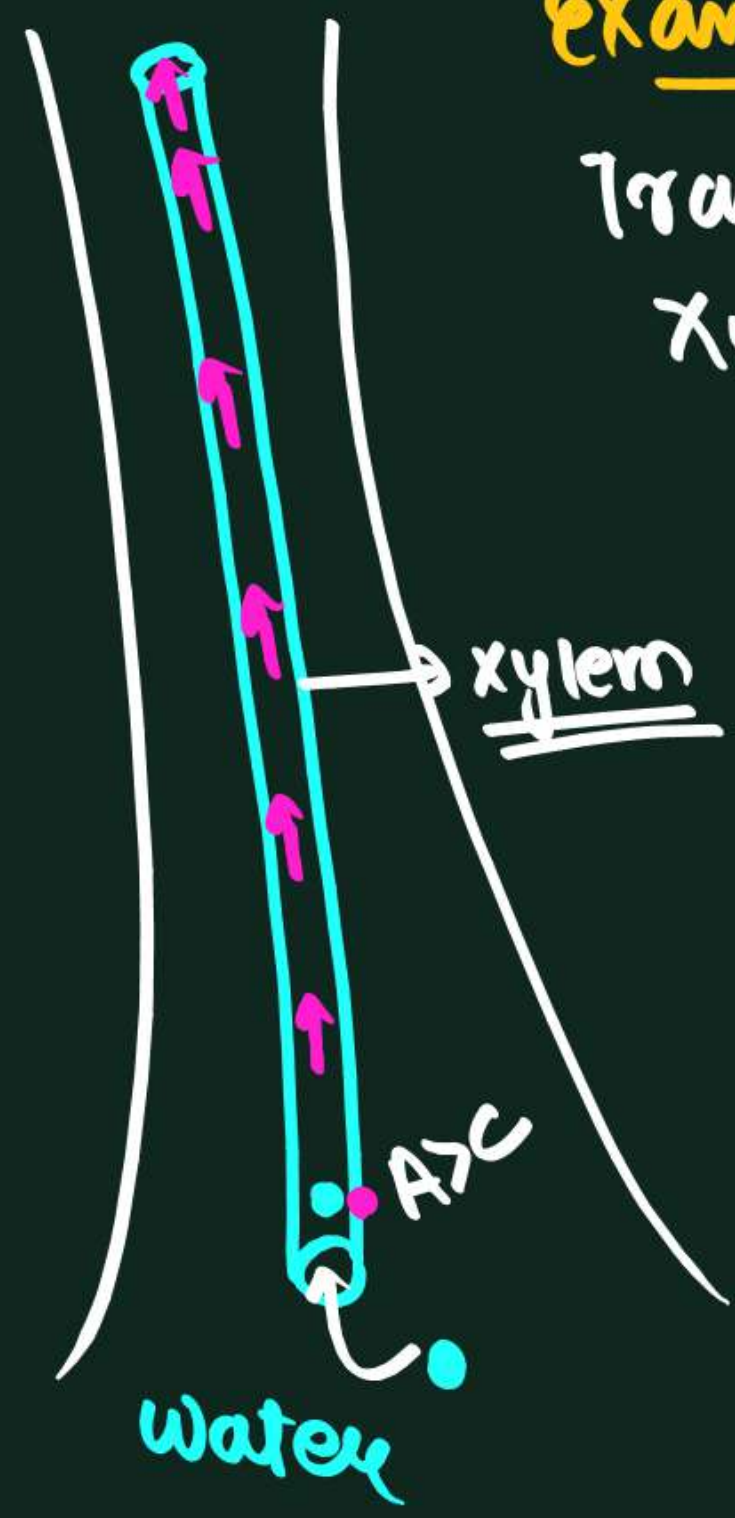
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The Rise of liquid (water) in capillarity tube when it is immersed in water is called capillary rise



Example:-

Transport of water through xylem in plants

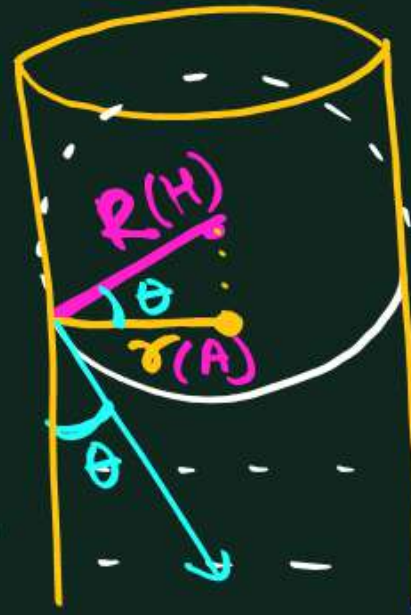


Q Derive an equation of Capillary Rise/Height in Capillary Rise?

3m / 5mark (5)
2026 2021
2025 2022

Sol

let us consider the container filled with (P) water. Immersed the capillary tube of radius (r) in water



Maths

$$P_B = P_0 + fgh$$

$$P_B - P_0 = fgh$$

$$P_B - P_0 = \epsilon P$$

$$\text{Excess pressure} = fgh$$

$$\# \cos \theta = \frac{r}{R}$$

$$\cos \theta = \frac{r}{R}$$

$$\frac{2T}{R} = fgh$$

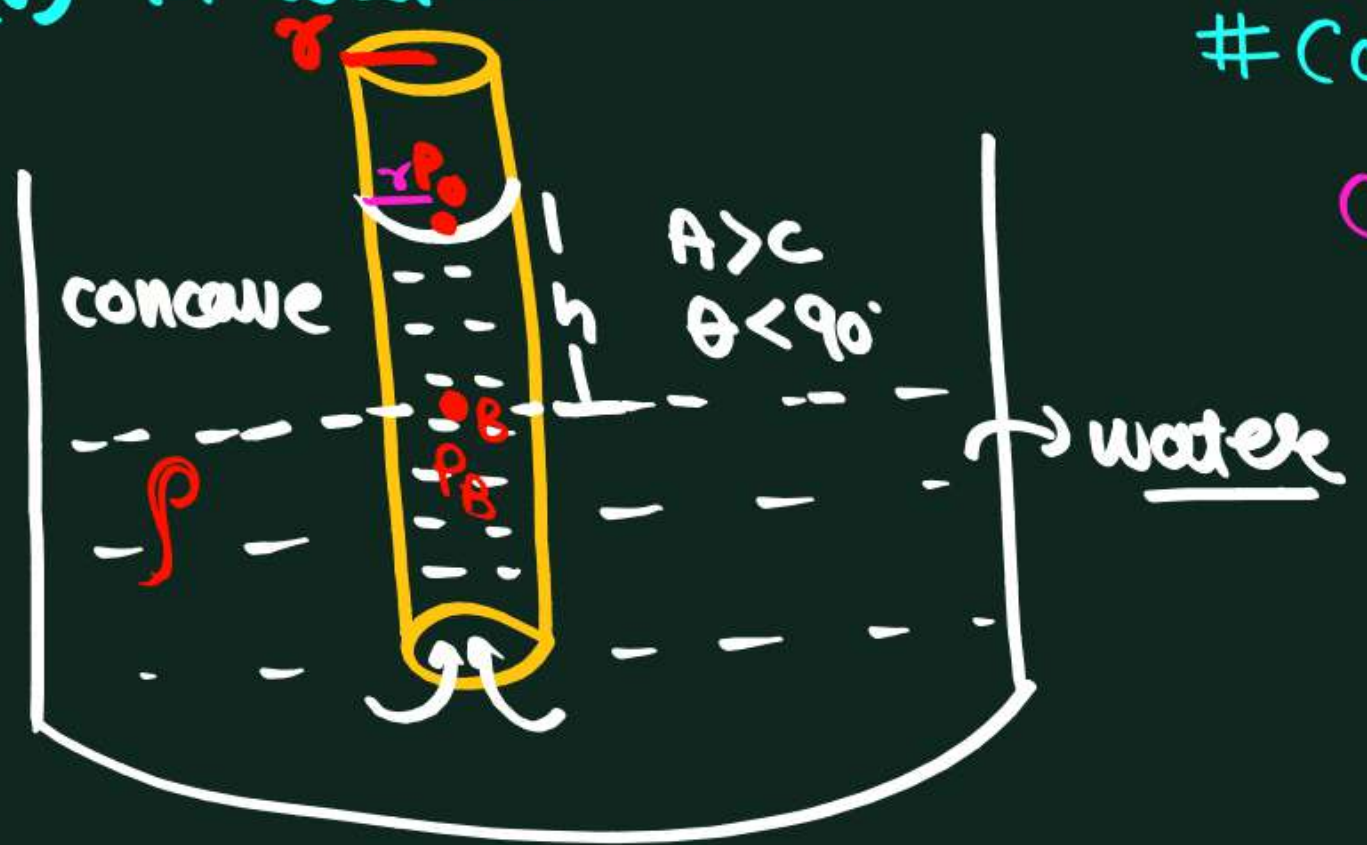
R → Radius of bubble

$$\frac{2T}{fgh} = h$$

r → radius of capillary tube

$$R = \frac{r}{\cos \theta}$$

$$\frac{2T}{fg \frac{r}{\cos \theta}} = h$$



$$h = \frac{2T}{\rho g \frac{r}{\cos \theta}}$$

A pink circle highlights the term $\frac{r}{\cos \theta}$ in the denominator, with a pink arrow pointing from it to the $2T$ in the numerator. A pink 'R' is written next to the circle.

6

$$h = \frac{2T \cos \theta}{\rho g r}$$

The entire equation is enclosed in a red rectangular box. The variable r in the denominator is highlighted in yellow.

h

Note

To prove $y = \theta$

Proof

$$\theta + \angle x = 90$$

$$\angle x = 90 - \theta$$

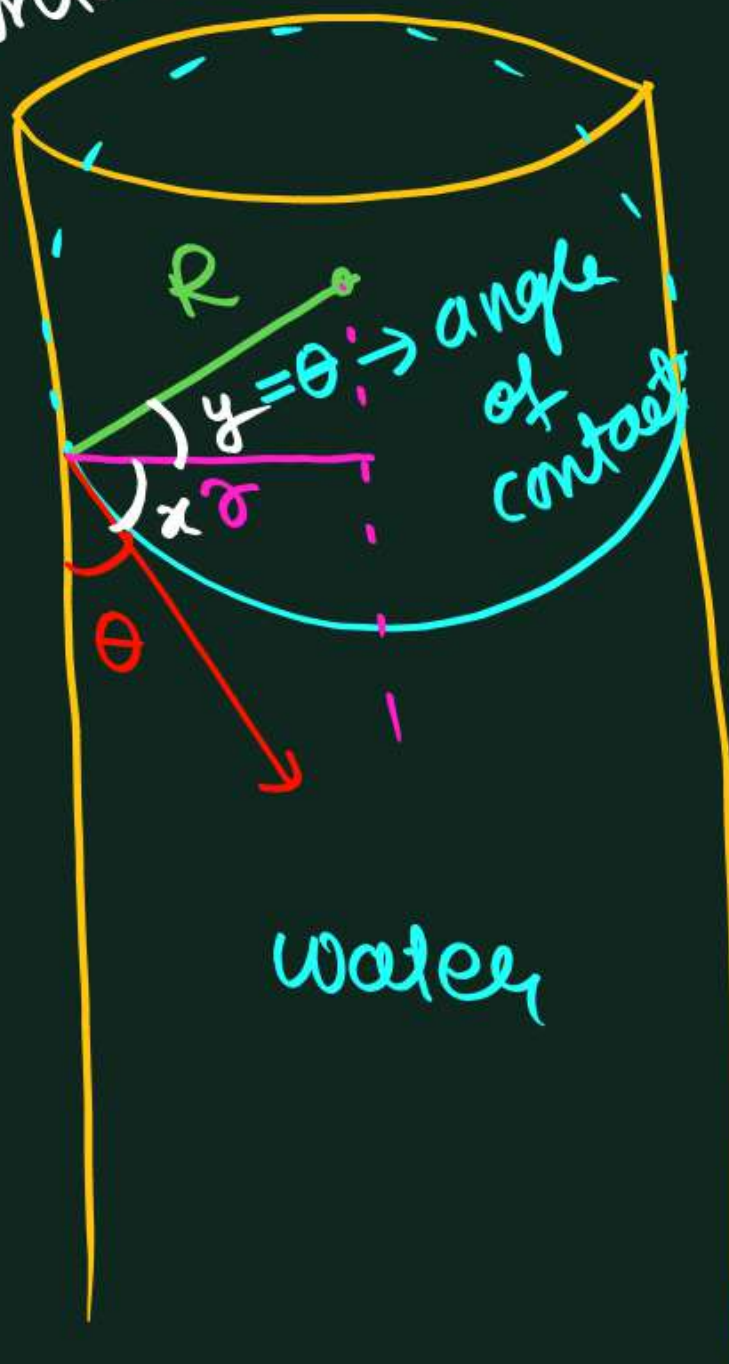
$$\Rightarrow \angle x + \angle y = 90$$

$$90 - \theta + \angle y = 90$$

$$\angle y = 90 - 90 + \theta$$

$$\angle y = \theta$$

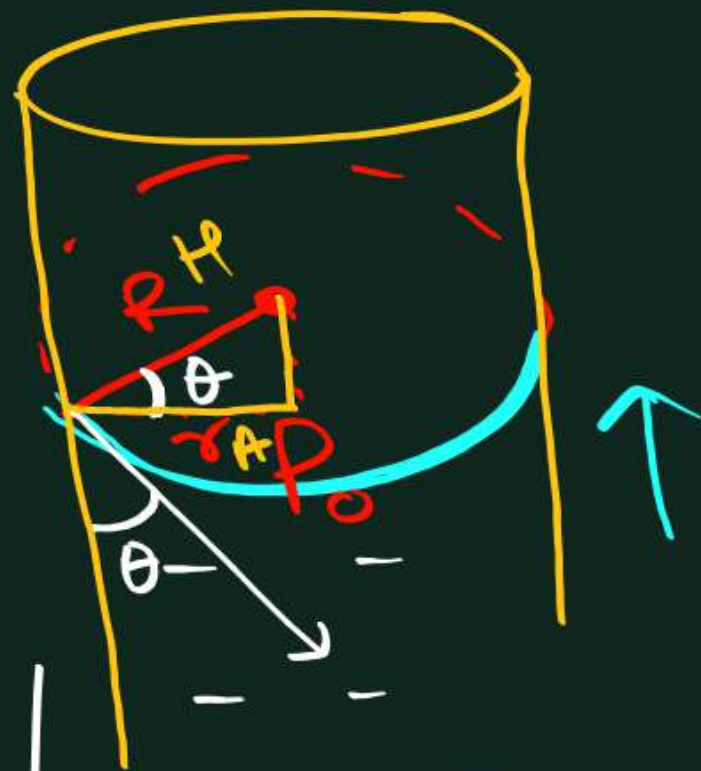
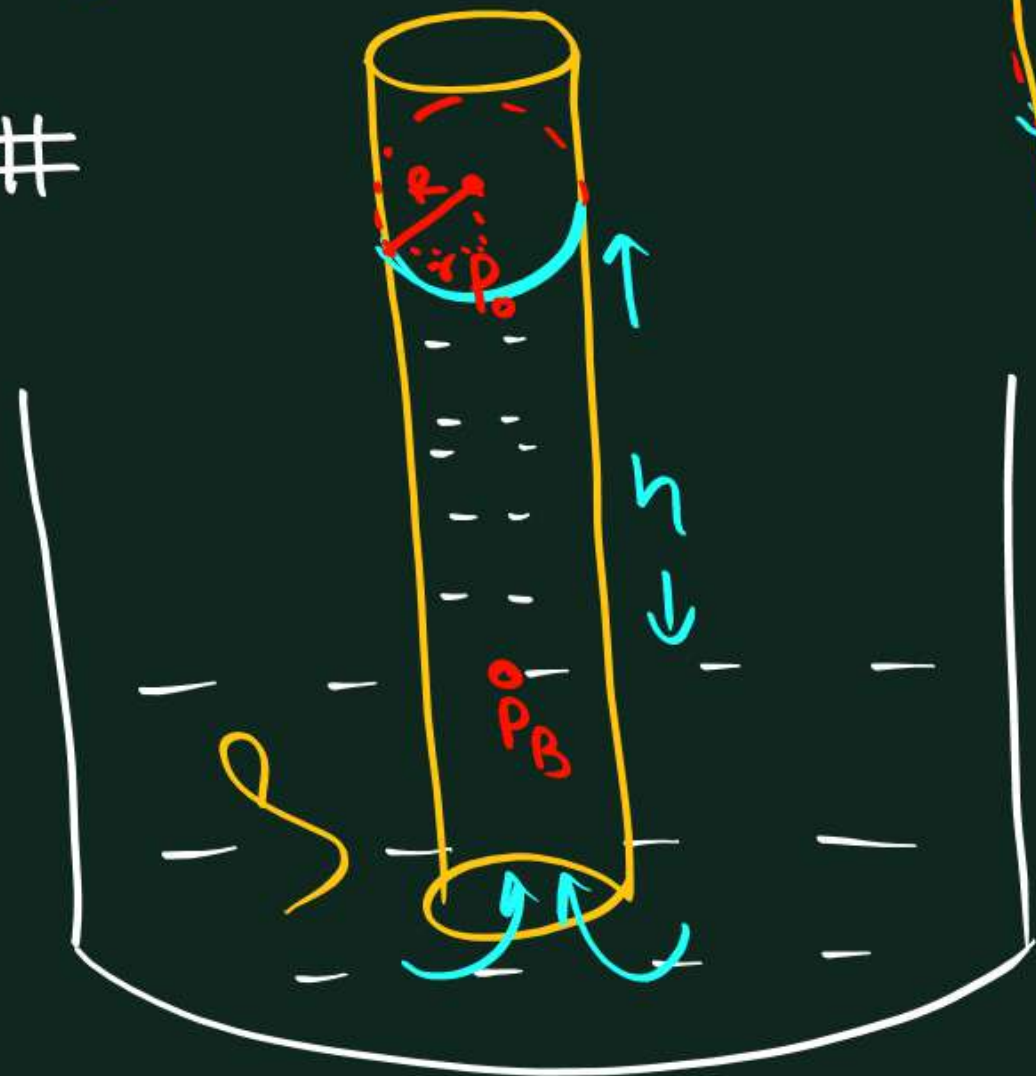
$\Rightarrow$ angle of contact



7

Rough

#



$$\cos \theta = \frac{h}{R}$$

$$\cos \theta = \frac{r}{R}$$

$$R = \frac{r}{\cos \theta}$$

Maths

$$P_B = P_0 + \rho g h$$

$$P_B - P_0 = \rho g h$$

$$\rho g h = \rho g h$$

$$\frac{2T}{R} = \rho g h$$

$$\frac{2T}{\rho g R} = h$$

$$\frac{2T}{\rho g \frac{r}{\cos \theta}} = h$$

$$\frac{2T \cos \theta}{\rho g r} = h$$

9

Feel of this formula

$$\uparrow h = \frac{2 \overset{\alpha}{I} \cos \theta}{\cancel{f} \cancel{g} \cancel{x}}$$

T↑↑	h↑↑
f↑↑	h↓↓
g↑↑	h↓↓
x↑↑	h↓↓

Q If you decrease the radius of capillary tube
the capillary height

Neet 2021 (10)

~~(a) Rise~~

(b) fall

(c) Remain same

(d) I donot know.

$$\uparrow h = \frac{2T \cos \theta}{\rho g r} \downarrow$$



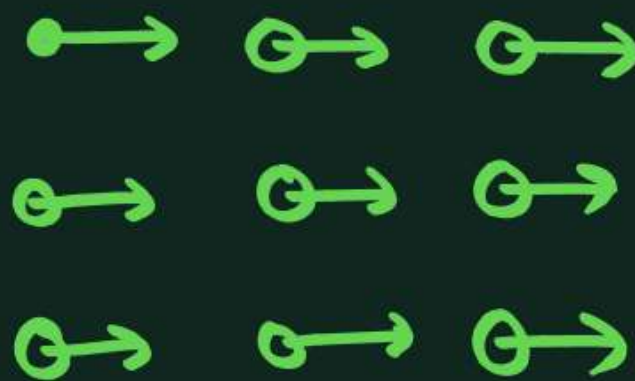
Critical velocity

2M - 2022 (JKBOSE)

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the velocity at which
Streamline flow /
laminar flow
change into
Turbulent
flow

Streamline flow



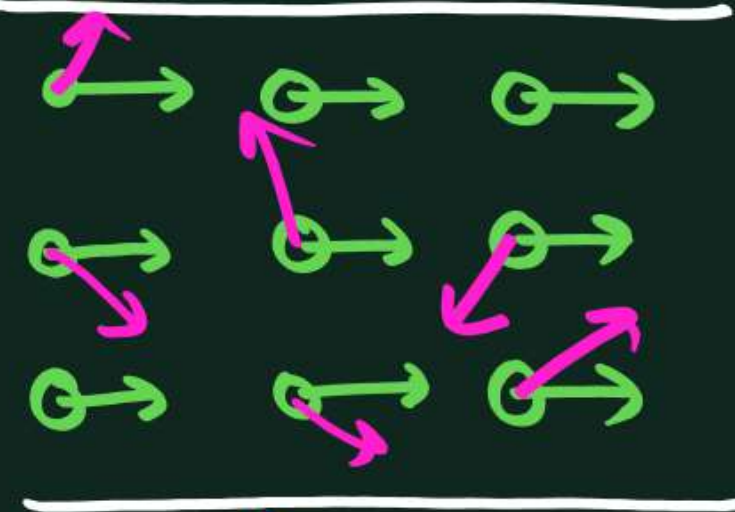
slow

$$V_s < V_c$$

$$9 < 10$$

$\Rightarrow$

Turbulent flow



Let

$$V_c = 10 \text{ m/s}$$

Fast

$$V_t > V_c$$

$$12 > 10$$

HOMework



$$x_h = \frac{2T \cos \theta}{\rho g}$$

CP

Handwritten notes.

CP



Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



Thank
You

